CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 9: WRITE THE PYTHON POGRAM TO IMPLEMENT TRAVELLING SALESMAN PROBLEM

Aim:

To write a Python program to implement the Travelling Salesman Problem (TSP) and find the minimum-cost tour that visits every city exactly once and returns to the starting city.

Algorithm:

1. Start the program.
2. Represent the distances between cities using a cost matrix.
3. Consider each city as a possible starting point.
4. Generate all possible routes that visit every city exactly once.
5. Calculate the total cost of each route, including the return to the starting city.
6. Find the route with the minimum total cost.
7. Display the optimal route and its minimum cost.
8. Stop the program.

PY8.py - C:/Users/navit/PY8.py (3.12.10)

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```
from itertools import permutations

def tsp(graph, start):
    cities = list(range(len(graph)))
    cities.remove(start)

    min_cost = float('inf')
    best_path = None

    for path in permutations(cities):
        route = (start,) + path + (start,)
        cost = 0

        for i in range(len(route) - 1):
            cost += graph[route[i]][route[i + 1]]

        if cost < min_cost:
            min_cost = cost
            best_path = route

    print("Optimal Route:")
    print(" -> ".join(map(str, best_path)))
    print("Minimum Cost:", min_cost)


graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

tsp(graph, 0)
```

IDLE Shell 3.12.10

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```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/navit/PY8.py ==========
Optimal Route:
0 -> 1 -> 3 -> 2 -> 0
Minimum Cost: 80
>>>
```

Result:

The Python program was executed successfully, and the optimal route and minimum travel cost were obtained.